



# Jorge Rodas-Silva

## Profesor e investigador en Ingeniería Informática

Profesor titular e investigador de la Universidad Estatal de Milagro (UNEMI) y Doctor en Ingeniería Informática por la Universidad de Sevilla, con reconocimiento Cum Laude y Distinción Internacional. Su trabajo integra inteligencia de negocios, analítica de datos educativos, transformación digital e ingeniería de software para impulsar la innovación en la educación superior. Cuenta con más de 15 años de trayectoria en investigación, docencia, innovación académica y gestión universitaria, y actualmente dirige la innovación de procesos académicos en UNEMI.

ORCID | Perfil científico | [jrodass@unemi.edu.ec](mailto:jrodass@unemi.edu.ec)

|               |            |          |                |
|---------------|------------|----------|----------------|
| 40            | 100        | 5        | 62%            |
| Publicaciones | Citaciones | Índice h | Acceso abierto |

### Formación académica

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| <b>Doctor en Ingeniería Informática</b><br>2019<br>Universidad de Sevilla · Sevilla, España<br>Cum Laude · Distinción Internacional            | <b>Magíster en Gerencia de Tecnologías de la Información</b><br>2011<br>Universidad Estatal de Milagro · Milagro, Ecuador<br>Mención de Honor · Mejor Graduado |
| <b>Ingeniero en Sistemas Computacionales</b><br>2010<br>Universidad Estatal de Milagro · Milagro, Ecuador<br>Mención de Honor · Mejor Graduado | <b>Business Intelligence and Datawarehouse (BIDWPC)</b><br>2013<br>Escuela Superior Politécnica del Litoral · Guayaquil, Ecuador<br>Certificación profesional  |

### Áreas de investigación

- Inteligencia de negocios y analítica de datos
  - Enseñanza de la programación y pensamiento computacional
  - Transformación digital en educación superior
- Minería de datos educativos y analítica del aprendizaje
  - Ingeniería de software y sistemas de recomendación
  - Inteligencia artificial aplicada a la educación

### Trayectoria académica y profesional

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|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2022 - Actualidad | <b>Director de Innovación de Procesos Académicos</b><br><br>Vicerrectorado de Formación Académica de Grado · Universidad Estatal de Milagro<br><a href="#">Gestión académica</a> |
| 2020 - 2024       | <b>Coordinador de la Maestría en Tecnologías de la Información</b><br><br>Universidad Estatal de Milagro<br><a href="#">Dirección de posgrado</a>                                |
| 2015 - Actualidad | <b>Profesor titular</b><br><br>Universidad Estatal de Milagro<br><a href="#">Inteligencia de negocios e ingeniería de software</a>                                               |

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|-------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| 2013 - 2014 | <b>Coordinador de Investigación</b><br><br>Facultad de Ciencias e Ingeniería · Universidad Estatal de Milagro<br><a href="#">Investigación</a> |
| 2008 - 2009 | <b>Programador sénior</b><br><br>Centro de Especialidades MegaSalud<br><a href="#">Industria</a>                                               |
| 2007 - 2008 | <b>Programador júnior</b><br><br>Gobierno Autónomo Descentralizado Simón Bolívar<br><a href="#">Industria</a>                                  |

## Publicaciones

Listado completo de 40 publicaciones, ordenadas de la más reciente a la más antigua.

|             |                                                                                                                                                                                                                                                                                                                                                                           |
|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>2026</b> | <b>Growth, concentration, and internationalisation of research output at a post-reform regional university in Ecuador: bibliometric evidence from Scopus (2016-2025)</b><br>Jorge Parraga-Alava, <b>Jorge Rodas-Silva</b><br>Scientometrics · 10.1007/s11192-026-05818-4                                                                                                  |
|             | <b>A graph-based semi-supervised approach for automatic text corpus emergency events labelling in X (Twitter)</b><br>Jorge Antonio Párraga Álava, <b>Jorge Rodas-Silva</b> , Jandry Franco, María-José Vélez<br>Big Data Research · 10.1016/j.bdr.2026.100630                                                                                                             |
|             | <b>Evaluating Zero-Shot and Few-Shot Vision-Language Models Against Supervised CNNs for Coffee Leaf Rust Classification</b><br><b>Jorge Rodas-Silva</b> , Jorge Antonio Párraga Álava<br>2026 8th Asia Conference on Machine Learning and Computing (ACMLC) · 10.1109/ACMLC70381.2026.11700499                                                                            |
|             | <b>Comparing Academic Performance in Online and Face-to-Face Object-Oriented Programming Courses: An Educational Data Mining Analysis</b><br><b>Jorge Rodas-Silva</b> , Jorge Antonio Párraga Álava<br>PROCEEDINGS OF 19TH IBERIAN CONFERENCE ON INFORMATION SYSTEMS AND TECHNOLOGIES, CISTI 2024, VOL 3 · 10.1007/978-3-032-12882-9_6                                    |
| <b>2025</b> | <b>Equity in academic evaluation and quality of university exams: a pattern analysis in higher education</b><br>J. Cobo, Fabricio Guevara-Viejó, <b>Jorge Rodas-Silva</b> , Cristian L. Vidal-Silva<br>Formacion Universitaria · 10.4067/S0718-50062025000600135                                                                                                          |
|             | <b>Computational Thinking and Academic Performance Across Different Instructional Modalities in Pre-University Courses: A Data-Driven Study</b><br>Jorge Antonio Párraga Álava, <b>Jorge Rodas-Silva</b><br>Engineering Reports · 10.1002/eng2.70203                                                                                                                      |
|             | <b>Bibliometric analysis of teaching strategies for programming courses in higher education, Análisis bibliométrico de estrategias de enseñanza para cursos de programación en educación superior</b><br>Mariuxi Vinueza-Morales, <b>Jorge Rodas-Silva</b> , Cristian L. Vidal-Silva, Erika Madariaga-García<br>Formacion Universitaria · 10.4067/s0718-50062025000400039 |
|             | <b>Deep Learning for Aphids Detection: A Comparative CNNs Study on Lemon Leaves Images</b><br>Jorge Antonio Párraga Álava, <b>Jorge Rodas-Silva</b> , Joseph Chávez Reyes, Gema Bravo Navarrete<br>EtcM 2025 9th Ecuador Technical Chapters Meeting · 10.1109/ETCM67548.2025.11304406                                                                                     |
|             | <b>Enhanced Helmet Detection using a CNN and Genetic Algorithm for Personal Protective Equipment Compliance</b><br><b>Jorge Rodas-Silva</b> , Jorge Antonio Párraga Álava<br>Conference Proceedings IEEE International Conference on Systems Man and Cybernetics · 10.1109/SMC58881.2025.11343106                                                                         |
|             | <b>Teaching programming in higher education: a bibliometric analysis of trends, technologies, and pedagogical approaches</b><br>Mariuxi Vinueza-Morales, <b>Jorge Rodas-Silva</b> , Cristian L. Vidal-Silva, Jorge Córdova-Morán, Edwin Cevallos-Ayón<br>Frontiers in Education · 10.3389/feduc.2025.1525917                                                              |
|             | <b>Teaching Programming in Higher Education: Analyzing Trends, Technologies, and Pedagogical Approaches Through a Bibliometric Lens</b><br>Mariuxi Vinueza-Morales, <b>Jorge Rodas-Silva</b> , Cristian L. Vidal-Silva<br>International Journal of Advanced Computer Science and Applications · 10.14569/IJACSA.2025.0160106                                              |
| <b>2024</b> | <b>An effective approach for identifying keywords as high-quality filters to get emergency-implicated Twitter Spanish data</b><br>Joel García-Arteaga, Jesús Zambrano-Zambrano, Jorge Antonio Párraga Álava, <b>Jorge Rodas-Silva</b><br>COMPUTER SPEECH AND LANGUAGE · 10.1016/j.CSL.2023.101579                                                                         |

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|             | <p><b>Applying Educational Programming Language-Based Learning to Enhance the Programming Skills of Non-ICT Engineering College Students</b><br/> <b>Jorge Rodas-Silva</b>, Jorge Antonio Párraga Álava, Mariuxi Vinueza-Morales, Ana Eva Chacón-Luna, J. Cobo<br/> Human Behavior and Emerging Technologies · 10.1155/2024/4918351</p>                                                                                             |
|             | <p><b>Enhancing Electric Power Consumption Prediction Using Machine Learning and Genetic Algorithms</b><br/> Jorge Antonio Párraga Álava, <b>Jorge Rodas-Silva</b><br/> EtcM 2024 8th Ecuador Technical Chapters Meeting · 10.1109/ETCM63562.2024.10746142</p>                                                                                                                                                                      |
| <b>2023</b> | <p><b>A Tuning Approach Using Genetic Algorithms for Emergency Incidents Classification in Social Media in Ecuador</b><br/> Joel García-Arteaga, Jesús Zambrano-Zambrano, Jorge Antonio Párraga Álava, <b>Jorge Rodas-Silva</b><br/> Communications in Computer and Information Science · 10.1007/978-3-031-32213-6_4</p>                                                                                                           |
|             | <p><b>Predicting Academic Performance of Low-Income Students in Public Ecuadorian Online Universities: An Educational Data Mining Approach</b><br/> <b>Jorge Rodas-Silva</b>, Jorge Antonio Párraga Álava<br/> Proceedings of the 12th International Conference on Data Science, Technology and Applications · 10.5220/0012086300003541</p>                                                                                         |
| <b>2022</b> | <p><b>Comparative Analysis of Methodologies and Technological Tools for Business Intelligence Processes Aimed at Decision-Making</b><br/> María José Guerrero García, <b>Jorge Rodas-Silva</b><br/> Informática y Sistemas · 10.33936/isrtic.v6i1.4522</p>                                                                                                                                                                          |
| <b>2021</b> | <p><b>Use of Web Applications for the Management of Veterinary Clinics and Their Impact on the Improvement of Administrative Processes</b><br/> Anthony Cedenó Ochoa, Axelman Catuto Murillo, <b>Jorge Rodas-Silva</b><br/> Ecuadorian Science Journal · 10.46480/esj.5.4.174</p>                                                                                                                                                   |
|             | <p><b>Opinion and Sentiment Analysis of Twitter Users during the 2021 Ecuador Presidential Election</b><br/> Jorge Antonio Párraga Álava, <b>Jorge Rodas-Silva</b>, Iván Quimi, Roberth Abel Alcivar Cevallos<br/> PROCEEDINGS OF THE 17TH INTERNATIONAL CONFERENCE ON WEB INFORMATION SYSTEMS AND TECHNOLOGIES (WEBIST) · 10.5220/0010651000003058</p>                                                                             |
| <b>2020</b> | <p><b>Teaching programming with MIT app inventor: A literature review, Enseñanza de programación mediante MIT App Inventor: Una revisión de literatura</b><br/> Mariuxi Vinueza-Morales, Ana Eva Chacón-Luna, <b>Jorge Rodas-Silva</b>, Héctor Serrano Mantilla<br/> Proceedings of the LACCEI international Multi-conference for Engineering, Education and Technology · 10.18687/LACCEI2020.1.1.49</p>                            |
| <b>2019</b> | <p><b>RESDEC: Online Management Tool for Implementation Components Selection in Software Product Lines Using Recommender Systems</b><br/> <b>Jorge Rodas-Silva</b>, José Á. Galindo, Jorge García-Gutiérrez, David Benavides<br/> 23RD INTERNATIONAL SYSTEMS AND SOFTWARE PRODUCT LINE CONFERENCE(SPLC 2019), VOL B · 10.1145/3307630.3342390</p>                                                                                   |
|             | <p><b>Automated analysis of two-layered feature models with feature attributes</b><br/> Michael Lettner, <b>Jorge Rodas-Silva</b>, José Á. Galindo, David Benavides<br/> Journal of Computer Languages · 10.1016/j.col.2019.01.005</p>                                                                                                                                                                                              |
|             | <p><b>Selection of Software Product Line Implementation Components Using Recommender Systems: An Application to Wordpress</b><br/> <b>Jorge Rodas-Silva</b>, José Á. Galindo, Jorge García-Gutiérrez, David Benavides<br/> IEEE ACCESS · 10.1109/ACCESS.2019.2918469</p>                                                                                                                                                            |
|             | <p><b>Using Alice software as a tool for programming learning: A literature review, El uso del software Alice como herramienta para el aprendizaje de programación: Una revisión de literatura</b><br/> Jorge Antonio Córdova Morán, <b>Jorge Rodas-Silva</b>, Mariuxi G. Vinueza Morales<br/> Proceedings of the LACCEI international Multi-conference for Engineering, Education and Technology · 10.18687/LACCEI2019.1.1.161</p> |
| <b>2018</b> | <p><b>Selection of Deployment Configurations Using Recommender Systems on Android</b><br/> <b>Jorge Rodas-Silva</b>, José A. Galindo, David Benavides, R. Soriano<br/> RISTI - Revista Ibérica de Sistemas e Tecnologias de Informação</p>                                                                                                                                                                                          |
|             | <p><b>RESDEC: A Prototype Tool for the Selection of Deployment Configurations Based on Recommender Systems</b><br/> <b>Jorge Rodas-Silva</b> et al.<br/> Actas de las 23rd Jornadas de Ingeniería del Software y Bases de Datos, JISBD 2018</p>                                                                                                                                                                                     |
| <b>2017</b> | <p><b>Programming with the SCRATCH + CARAMBA tool. A significant learning experience, Programación con la herramienta SCRATCH +CARAMBA. Una experiencia de aprendizaje significativo</b><br/> J. Cobo, David Benavides, Mayra D'Armas, Mariuxi Vinueza, <b>Jorge Rodas-Silva</b><br/> Proceedings of the LACCEI international Multi-conference for Engineering, Education and Technology · 10.18687/LACCEI2017.1.1.268</p>          |
| <b>2016</b> | <p><b>Using Attributed Feature Models for Testing Highly Configurable Systems: First Steps</b><br/> Mariuxi Vinueza, <b>Jorge Rodas-Silva</b>, José Ángel Galindo Duarte, David Benavides<br/> Actas de las 21st Jornadas de Ingeniería y Bases de Datos, JISBD 2016</p>                                                                                                                                                            |
|             | <p><b>The use of feature models with attributes for testing variability intensive systems: First steps</b><br/> <b>Jorge Rodas-Silva</b> et al.<br/> Actas De Las 21st Jornadas De Ingenier&amp;amp;amp;amp;amp;amp;#65533 A Del Software Y Bases De Datos Jisbd 2016</p>                                                                                                                                                           |

**Towards the Use of Recommender Systems in Highly Configurable Systems**  
**Jorge Rodas-Silva**, Olivares, Javier, José Ángel Galindo Duarte, David Benavides  
Actas de las 21st Jornadas de Ingenier&amp;amp;amp;amp;amp;#65533;a del Software y Bases de Datos, JISBD 2016

**Towards the use of recommender systems in variability intensive systems**  
**Jorge Rodas-Silva** et al.  
Actas De Las 21st Jornadas De Ingenier Amp Amp Amp Amp Amp 65533 A Del Software Y Bases De Datos Jisbd 2016

**Body Bias usage in UTBB FDSOI designs: A parametric exploration approach**  
**Jorge Rodas-Silva** et al.  
SOLID-STATE ELECTRONICS · 10.1016/j.SSE.2015.11.019

**2015** **Business Intelligence System and Its Impact on Decision-Making in Telecommunications Companies in La Troncal, Ecuador**  
**Jorge Luis Rodas-Silva**, Manuel Guillermo Rodríguez-López, Jesennia Cárdenas-Cobo  
Copérnico

**Towards testing variability intensive systems using user reviews**  
**Jorge Rodas-Silva**, David Méndez-Acuña, José A. Galindo, David Benavides, Jesennia Cardenas  
2015 10TH COMPUTING COLOMBIAN CONFERENCE (10CCC) · 10.1109/ColumbianCC.2015.7333410

**Strategic Planning Through ICT in Rural Decentralized Autonomous Governments of Milagro Canton**  
Mariuxi Geovanna Vinueza Morales, **Jorge Rodas-Silva**, Ana Chacón Luna  
Ciencia UNEMI · 10.29076/issn.2528-7737vol8iss15.2015pp40-49p

**Standards That Contribute to the Development and Delivery of High-Quality Software Products**  
Ana Chacón Luna, **Jorge Luis Rodas-Silva**, Mariuxi Vinueza Morales  
Ciencia UNEMI · 10.29076/issn.2528-7737vol8iss13.2015pp90-99p

**Evaluation and Mitigation of Aging Effects on a Digital On-Chip Voltage and Temperature Sensor**  
**Jorge Rodas-Silva** et al.  
PROCEEDINGS 2015 25TH INTERNATIONAL WORKSHOP ON POWER AND TIMING MODELING, OPTIMIZATION AND SIMULATION

**2014** **Digital Management Systems to Improve Academic Processes in Educational Institutions**  
**Jorge Rodas-Silva**, Jesennia Cárdenas-Cobo  
Universidad, Ciencia y Tecnología

**E-Commerce: A Perspective from SMEs for Developing Strategies to Promote Economic and Business Growth in the City of Milagro**  
**Jorge Luis Rodas-Silva**, Ana Eva Chacón Luna, Mariuxi Geovanna Vinueza Morales  
ECA Sinergia

**2012** **Study to Determine the Use and Application of ICT in Teaching and Learning Processes by Educators in Milagro and Neighboring Cantons**  
**Jorge Luis Rodas-Silva**  
Ciencia UNEMI · 10.29076/issn.2528-7737vol5iss8.2012pp79-92p

Reconocimientos

|      |                                                                 |                                                                     |
|------|-----------------------------------------------------------------|---------------------------------------------------------------------|
| 2019 | <b>Doctorado Cum Laude con Distinción Internacional</b>         | Decisión unánime · Universidad de Sevilla                           |
| 2011 | <b>Mejor graduado de la Maestría en Gerencia de TI</b>          | Mención de Honor · Primera cohorte · Universidad Estatal de Milagro |
| 2006 | <b>Mejor graduado de Ingeniería en Sistemas Computacionales</b> | Mención de Honor · Universidad Estatal de Milagro                   |

Indicadores actualizados: 2026-09-29. Fuentes: ORCID, Scopus, OpenAlex, Latindex, Crossref.